

DATA WAREHOUSING AND DATA MINING

DAY - 3 LAB PROGRAMS

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Question-1

Consider the dataset and perform the Apriori Algorithm and FP algorithm with support = 3 and confidence = 50%.

Aim

To find frequent itemsets and generate association rules using the **Apriori** and **FP-Growth** algorithms.

Steps:

1. Open WEKA → Explorer.
2. Load market_basket.arff.
3. Associate → Apriori → Support 0.30, Confidence 0.50 → Start.
4. Associate → FP-Growth → Support 0.30, Confidence 0.50 → Start.
5. Record and compare the results.

Output:

```
23:05:52 - Apriori

=== Run information ===

Scheme:      weka.associations.Apriori -N 10 -T 0 -C 0.5 -D 0.01 -U 0.3 -M 0.3 -S -1.0 -c -1
Relation:     market_basket
Instances:    10
Attributes:   4
              Milk
              Bread
              Butter
              Eggs

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.35 (3 instances)
Minimum metric <confidence>: 0.5
Number of cycles performed: 65

Generated sets of large itemsets:

Size of set of large itemsets L(1): 7

Size of set of large itemsets L(2): 12

Size of set of large itemsets L(3): 6

Best rules found:

1. Butter=no 3 ==> Milk=yes 3    <conf:(1)> lift:(1.43) lev:(0.09) [0] conv:(0.9)
2. Milk=no 3 ==> Bread=yes 3    <conf:(1)> lift:(1.11) lev:(0.03) [0] conv:(0.3)
3. Milk=no 3 ==> Butter=yes 3   <conf:(1)> lift:(1.43) lev:(0.09) [0] conv:(0.9)
4. Milk=yes Eggs=no 3 ==> Bread=yes 3 <conf:(1)> lift:(1.11) lev:(0.03) [0] conv:(0.3)
5. Milk=no Butter=yes 3 ==> Bread=yes 3 <conf:(1)> lift:(1.11) lev:(0.03) [0] conv:(0.3)
6. Milk=no Bread=yes 3 ==> Butter=yes 3 <conf:(1)> lift:(1.43) lev:(0.09) [0] conv:(0.9)
7. Milk=no 3 ==> Bread=yes Butter=yes 3 <conf:(1)> lift:(1.43) lev:(0.09) [0] conv:(0.9)
8. Butter=yes Eggs=yes 3 ==> Bread=yes 3 <conf:(1)> lift:(1.11) lev:(0.03) [0] conv:(0.3)
9. Bread=yes Eggs=yes 4 ==> Milk=yes 3 <conf:(0.75)> lift:(1.07) lev:(0.02) [0] conv:(0.6)
10. Milk=yes Eggs=yes 4 ==> Bread=yes 3 <conf:(0.75)> lift:(0.83) lev:(-0.06) [0] conv:(0.2)
```

Associator

Choose **FPGrowth** -P 1 -I -1 -N 10 -T 0 -C 0.5 -D 0.01 -U 0.3 -M 0.3

Start Stop

Result list (right-click for ...)

23:05:52 - Apriori

00:21:30 - FPGrowth

Associator output

```

=== Run information ===

Scheme:      weka.associations.FPGrowth -P 1 -I -1 -N 10 -T 0 -C 0.5 -D 0.01 -U 0.3 -M 0.3
Relation:    market_basket
Instances:   10
Attributes:  4
              Milk
              Bread
              Butter
              Eggs

=== Associator model (full training set) ===

FPGrowth found 8 rules (displaying top 8)

1. [Butter=yes, Eggs=yes]: 3 ==> [Bread=yes]: 3    <conf:(1)> lift:(1.11) lev:(0.03) conv:(0.3)
2. [Bread=yes, Eggs=yes]: 4 ==> [Milk=yes]: 3    <conf:(0.75)> lift:(1.07) lev:(0.02) conv:(0.6)
3. [Milk=yes, Eggs=yes]: 4 ==> [Bread=yes]: 3    <conf:(0.75)> lift:(0.83) lev:(-0.06) conv:(0.2)
4. [Bread=yes, Eggs=yes]: 4 ==> [Butter=yes]: 3    <conf:(0.75)> lift:(1.07) lev:(0.02) conv:(0.6)
5. [Eggs=yes]: 5 ==> [Butter=yes]: 3    <conf:(0.6)> lift:(0.86) lev:(-0.05) conv:(0.5)
6. [Eggs=yes]: 5 ==> [Bread=yes, Milk=yes]: 3    <conf:(0.6)> lift:(1) lev:(0) conv:(0.67)
7. [Eggs=yes]: 5 ==> [Bread=yes, Butter=yes]: 3    <conf:(0.6)> lift:(0.86) lev:(-0.05) conv:(0.5)
8. [Bread=yes, Milk=yes]: 6 ==> [Eggs=yes]: 3    <conf:(0.5)> lift:(1) lev:(0) conv:(0.75)

```

Question 2

Consider the market basket transactions shown in the above table.

- (a) What is the maximum number of association rules that can be extracted from this data, including rules that have zero support?
- (b) What is the maximum size of frequent itemsets that can be extracted, assuming minimum support > 0 ?

Aim

To determine the **maximum number of association rules** and the **maximum size of frequent itemsets** from the given market-basket dataset.

Steps

1. Identify the number of distinct items (n) in the dataset.
2. Calculate maximum association rules using:
3. Find the transaction containing the maximum number of items.
4. That number is the maximum possible size of a frequent itemset.
5. Verify using WEKA → Explorer → Associate → Apriori if required.

Output:

(a) Maximum number of association rules

$$3^n - 2^{n+1} + 1$$

$$\begin{aligned}
 \text{Substitute } (n = 6): & 3^6 - 2^7 + 1 \\
 &= 729 - 128 + 1 \\
 &= 602
 \end{aligned}$$

Maximum number of association rules = 602

(b) Maximum size of frequent itemsets = 4

```
00:29:55 - Apriori
Cookies
=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.55 (5 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 9

Generated sets of large itemsets:

Size of set of large itemsets L(1): 9

Size of set of large itemsets L(2): 5

Size of set of large itemsets L(3): 1

Best rules found:

1. Bread=yes 5 ==> Beer=no 5    <conf:(1)> lift:(1.67) lev:(0.2) [2] conv:(2)
2. Butter=yes 5 ==> Beer=no 5    <conf:(1)> lift:(1.67) lev:(0.2) [2] conv:(2)
3. Butter=yes 5 ==> Bread=yes 5   <conf:(1)> lift:(2) lev:(0.25) [2] conv:(2.5)
4. Bread=yes 5 ==> Butter=yes 5   <conf:(1)> lift:(2) lev:(0.25) [2] conv:(2.5)
5. Butter=no 5 ==> Bread=no 5     <conf:(1)> lift:(2) lev:(0.25) [2] conv:(2.5)
6. Bread=no 5 ==> Butter=no 5     <conf:(1)> lift:(2) lev:(0.25) [2] conv:(2.5)
7. Bread=yes Butter=yes 5 ==> Beer=no 5 <conf:(1)> lift:(1.67) lev:(0.2) [2] conv:(2)
8. Beer=no Butter=yes 5 ==> Bread=yes 5 <conf:(1)> lift:(2) lev:(0.25) [2] conv:(2.5)
9. Beer=no Bread=yes 5 ==> Butter=yes 5 <conf:(1)> lift:(2) lev:(0.25) [2] conv:(2.5)
10. Butter=yes 5 ==> Beer=no Bread=yes 5 <conf:(1)> lift:(2) lev:(0.25) [2] conv:(2.5)
```

Question 3 — Bayes Classification and Decision Tree

Perform Bayes classification and Decision Tree classification using training and test data.

Aim

To classify the given data using **Naive Bayes** and **Decision Tree (J48)** algorithms in WEKA using training and test data.

Steps

1. Open WEKA → Explorer.
2. Go to Preprocess → Open file and load the dataset.
3. Go to Classify.
4. Select Choose → bayes → NaiveBayes.
5. Select Use training set and click Start.
6. Record the classification accuracy.
7. Select Choose → trees → J48.
8. Select Use training set and click Start.
9. For testing, select Supplied test set → Set and load the test dataset.

10. Run both NaiveBayes and J48 and compare their accuracy.

Output:

```
00:39:37 - bayes.NaiveBayes

Time taken to build model: 0 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.01 seconds

=== Summary ===

Correctly Classified Instances      13          92.8571 %
Incorrectly Classified Instances    1           7.1429 %
Kappa statistic                    0.8372
Mean absolute error                 0.2917
Root mean squared error             0.3392
Relative absolute error             62.8233 %
Root relative squared error        70.7422 %
Total Number of Instances          14

=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
          1.000    0.200    0.900     1.000    0.947     0.849    0.922    0.947    yes
          0.800    0.000    1.000     0.800    0.889     0.849    0.911    0.911    no
Weighted Avg.   0.929    0.129    0.936     0.929    0.926     0.849    0.918    0.934

=== Confusion Matrix ===

 a b   <-- classified as
 9 0 | a = yes
 1 4 | b = no
```

```
00:40:42 - trees.J48

Time taken to build model: 0.03 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances      14          100 %
Incorrectly Classified Instances    0           0 %
Kappa statistic                    1
Mean absolute error                 0
Root mean squared error             0
Relative absolute error             0 %
Root relative squared error         0 %
Total Number of Instances          14

=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
          1.000    0.000    1.000     1.000    1.000     1.000    1.000    1.000    yes
          1.000    0.000    1.000     1.000    1.000     1.000    1.000    1.000    no
Weighted Avg.   1.000    0.000    1.000     1.000    1.000     1.000    1.000    1.000

=== Confusion Matrix ===

 a b   <-- classified as
 9 0 | a = yes
 0 5 | b = no
```

Question – 4:

Analyze the diabetes.csv dataset to study the diabetes trend among different age groups using Linear Regression and Multiple Regression in R.

Aim

To analyze the relationship between age and diabetes, and to determine how age along with other health factors (Glucose, BMI, Blood Pressure, and Insulin) influences diabetes outcome using regression techniques in R.

Code:

```
diabetes <- read.csv("C:\\Users\\balaj\\Downloads\\diabetes.csv")
```

```
head(diabetes)
```

```
str(diabetes)
```

```
summary(diabetes)
```

```
linear_model <- lm(Outcome ~ Age, data = diabetes)
```

```
cat("Linear Regression Result:\n")
```

```
summary(linear_model)
```

```
plot(diabetes$Age, diabetes$Outcome,
```

```
      main = "Age vs Diabetes Outcome",
```

```
      xlab = "Age",
```

```
      ylab = "Diabetes Outcome",
```

```
      pch = 19,
```

```
      col = "blue")
```

```
abline(linear_model, col = "red", lwd = 2)
```

```
multiple_model <- lm(Outcome ~ Age + Glucose + BMI +
```

```
      BloodPressure + Insulin,
```

```
data = diabetes)
```

```
cat("Multiple Regression Result:\n")
```

```
summary(multiple_model)
```

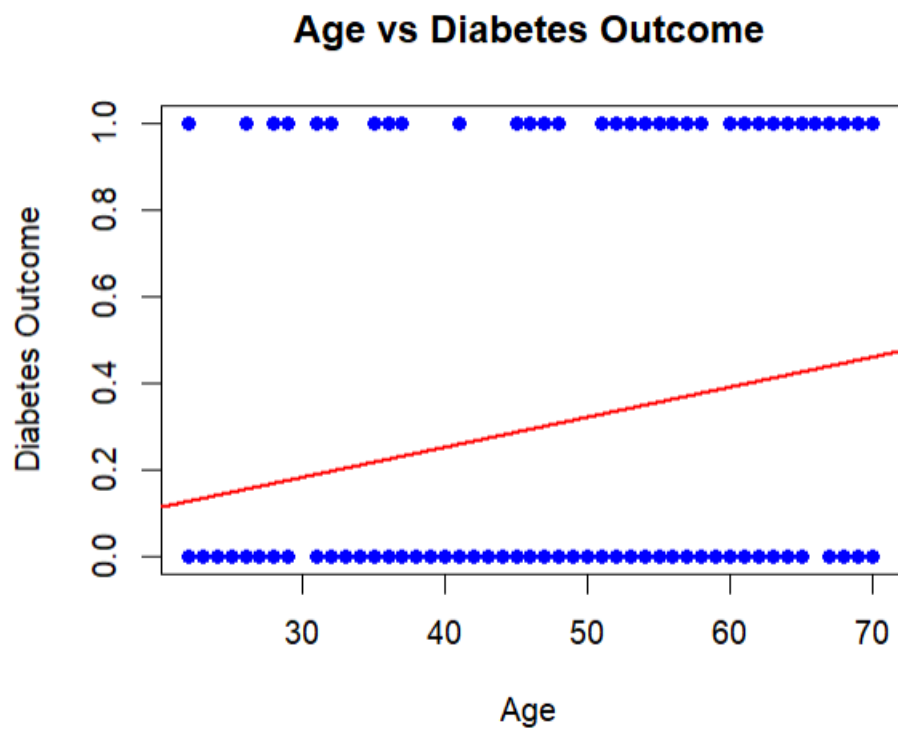
```
diabetes$Predicted_Outcome <- predict(multiple_model)
```

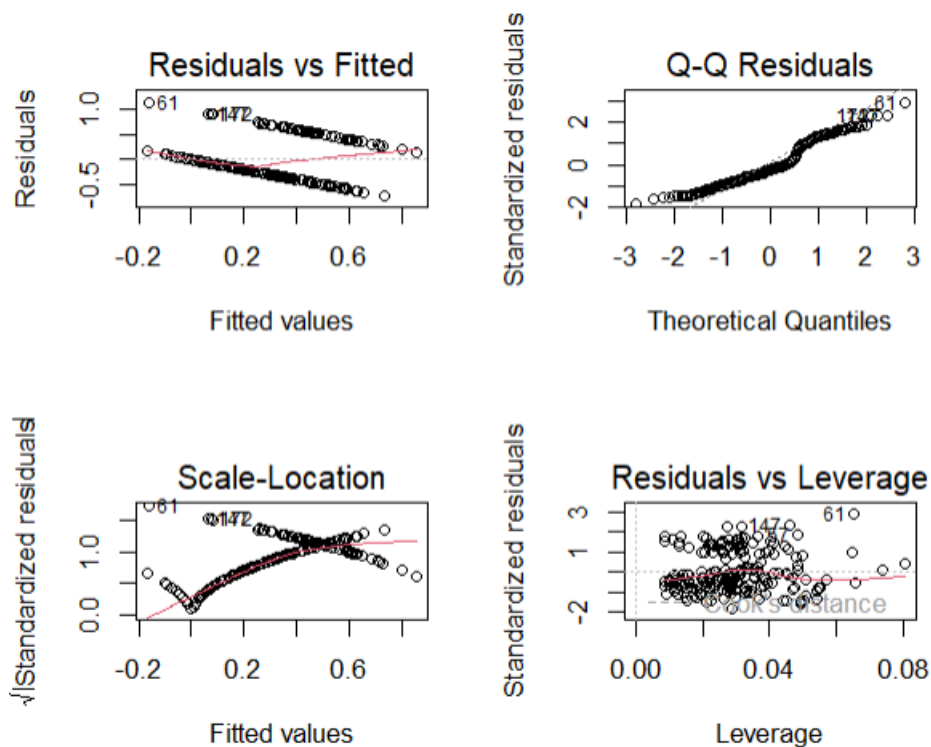
```
head(diabetes)
```

```
par(mfrow = c(2,2))
```

```
plot(multiple_model)
```

Output:





Question – 5:

Implement Apriori and FP-Growth algorithms in WEKA for the given transaction database with minimum support = 50% (2 transactions) and minimum confidence = 80%, and find all frequent itemsets and association rules.

Aim

To discover frequent itemsets and generate association rules from a transaction database using Apriori and FP-Growth algorithms in WEKA.

Steps (WEKA)

1. Open WEKA Explorer.
2. Prepare the transaction dataset in ARFF or CSV format.
3. Click Preprocess and load the dataset.
4. Go to the Associate tab.
5. Select Apriori as the association rule mining algorithm.
6. Set Minimum Support = 0.5 and Minimum Confidence = 0.8.
7. Click Start to generate frequent itemsets and association rules.
8. Note the output results.
9. Again, in the Associate tab, select FPGrowth.
10. Use the same support and confidence values.

11. Click Start to generate the FP-Growth frequent itemsets and rules.

12. Compare the results obtained from Apriori and FP-Growth.

Output:

```
08:53:56 - Apriori
A=f 4
U=f 4
I=f 4

Size of set of large itemsets L(2): 6

Large Itemsets L(2):
K=t E=t 4
K=t D=f 4
K=t A=f 4
K=t U=f 4
K=t I=f 4
E=t U=f 4

Size of set of large itemsets L(3): 1

Large Itemsets L(3):
K=t E=t U=f 4

Best rules found:

1. E=t 4 ==> K=t 4    <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
2. D=f 4 ==> K=t 4    <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
3. A=f 4 ==> K=t 4    <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
4. U=f 4 ==> K=t 4    <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
5. I=f 4 ==> K=t 4    <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
6. U=f 4 ==> E=t 4    <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
7. E=t 4 ==> U=f 4    <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
8. E=t U=f 4 ==> K=t 4    <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
9. K=t U=f 4 ==> E=t 4    <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
10. K=t E=t 4 ==> U=f 4    <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
```

```
08:56:05 - FPGrowth

=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 20 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.5
Relation:    transactions
Instances:    5
Attributes:   11
              M
              O
              N
              K
              E
              Y
              D
              A
              U
              C
              I

=== Associator model (full training set) ===

FPGrowth found 7 rules (displaying top 7)

1. [C=f]: 3 ==> [U=f]: 3    <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
2. [C=f]: 3 ==> [I=f]: 3    <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
3. [N=f]: 3 ==> [D=f]: 3    <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
4. [U=f, I=f]: 3 ==> [C=f]: 3    <conf:(1)> lift:(1.67) lev:(0.24) conv:(1.2)
5. [C=f]: 3 ==> [U=f, I=f]: 3    <conf:(1)> lift:(1.67) lev:(0.24) conv:(1.2)
6. [U=f, C=f]: 3 ==> [I=f]: 3    <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
7. [I=f, C=f]: 3 ==> [U=f]: 3    <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
```


Question - 6

Predict categorical data using the Decision Tree Algorithm in WEKA using any dataset and observe the Preprocess, Tree, and Logistic results.

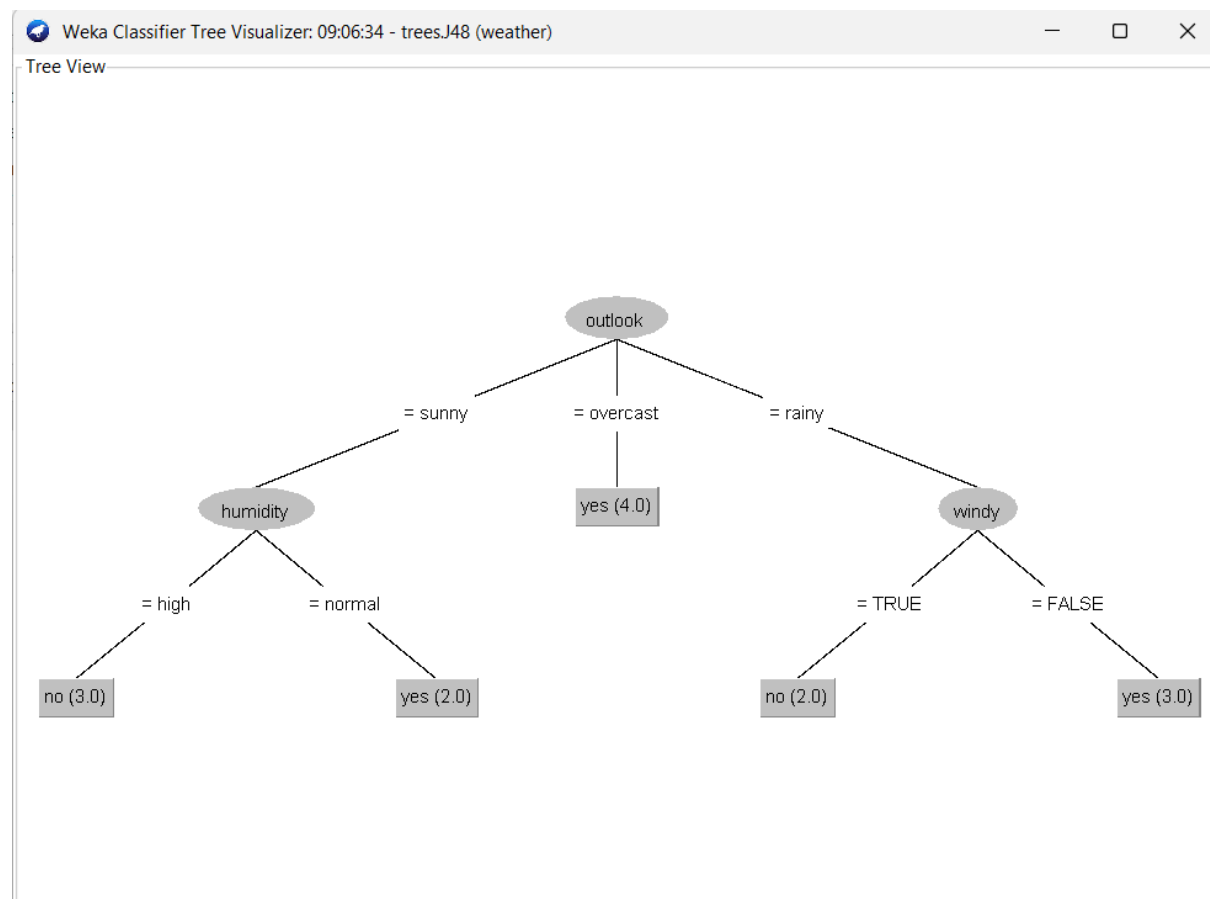
Aim

To classify categorical data using the J48 Decision Tree algorithm and compare it with Logistic classification in WEKA.

Steps:

1. Open WEKA and select Explorer.
2. Go to the Preprocess tab.
3. Click Open file and load weather.nominal.arff.
4. Verify that the class attribute is play. Open the Classify tab.
5. Click Choose → trees → J48 to select the Decision Tree algorithm.
6. Under Test options, select Use training set.
7. Click Start to generate the decision tree and view the classification results.
8. Again, click Choose → functions → Logistic.
9. Keep Use training set selected.
10. Click Start to build the Logistic classifier and view its results.
11. Compare the outputs of J48 and Logistic classifiers.

Output:



```
09:06:34 - trees.J48

Time taken to build model: 0.03 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances      14          100   %
Incorrectly Classified Instances    0           0   %
Kappa statistic                     1
Mean absolute error                 0
Root mean squared error             0
Relative absolute error             0   %
Root relative squared error         0   %
Total Number of Instances          14

=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
          1.000    0.000    1.000    1.000    1.000     1.000    1.000    1.000    yes
          1.000    0.000    1.000    1.000    1.000     1.000    1.000    1.000    no
Weighted Avg.   1.000    0.000    1.000    1.000    1.000     1.000    1.000    1.000

=== Confusion Matrix ===

a b   <-- classified as
9 0 | a = yes
0 0 | b = no
```

```
09:08:54 - functions.Logistic

Time taken to build model: 0.02 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances      14          100   %
Incorrectly Classified Instances    0           0   %
Kappa statistic                     1
Mean absolute error                 0
Root mean squared error             0
Relative absolute error             0.0002 %
Root relative squared error         0.0008 %
Total Number of Instances          14

=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
          1.000    0.000    1.000    1.000    1.000     1.000    1.000    1.000    yes
          1.000    0.000    1.000    1.000    1.000     1.000    1.000    1.000    no
Weighted Avg.   1.000    0.000    1.000    1.000    1.000     1.000    1.000    1.000

=== Confusion Matrix ===

a b   <-- classified as
9 0 | a = yes
0 5 | b = no
```

Question 7 - Apriori & FP-Growth in WEKA

Create the dataset in ARFF format and find the frequent itemsets and association rules using Apriori and FP-Growth algorithms in WEKA. Assume Minimum Support = 33.33% and Minimum Confidence = 60%.

Aim

To generate frequent itemsets and association rules from transaction data using Apriori and FP-Growth algorithms in WEKA.

Steps

1. Open WEKA Explorer.
2. Go to the Preprocess tab.
3. Load the transaction ARFF dataset.
4. Open the Associate tab.
5. Click Choose → Apriori.
6. Set Minimum Support = 0.333 and Minimum Confidence = 0.6.
7. Click Start to generate frequent itemsets and association rules.
8. Save or note the output.
9. Again click Choose → FPGrowth.
10. Set the same support (0.333) and confidence (0.6).
11. Click Start.
12. Compare the rules generated by Apriori and FP-Growth.

Output:

```
Choose Apriori -N 10 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.333 -S -1.0 -c -1

Start Stop
Result list (right-click for ...
09:18:08 - Apriori

Associator output
Attributes:
HotDogs
Buns
Ketchup
Coke
Chips

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.55 (3 instances)
Minimum metric <confidence>: 0.6
Number of cycles performed: 9

Generated sets of large itemsets:

Size of set of large itemsets L(1): 6
Size of set of large itemsets L(2): 7
Size of set of large itemsets L(3): 4
Size of set of large itemsets L(4): 1

Best rules found:

1. Chips=t 4 ==> Buns=f 4 <conf:(1)> lift:(1.5) lev:(0.22) [1] conv:(1.33)
2. Buns=f 4 ==> Chips=t 4 <conf:(1)> lift:(1.5) lev:(0.22) [1] conv:(1.33)
3. Coke=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
4. Coke=t 3 ==> Ketchup=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
5. Coke=t 3 ==> Chips=t 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
6. Ketchup=f Coke=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
7. Buns=f Coke=t 3 ==> Ketchup=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
8. Buns=f Ketchup=f 3 ==> Coke=t 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)
9. Coke=t 3 ==> Buns=f Ketchup=f 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)
10. Ketchup=f Chips=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

Status
OK
```

```
09:19:36 - FPGrowth

=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.333
Relation:     transactions
Instances:    6
Attributes:   5
              HotDogs
              Buns
              Ketchup
              Coke
              Chips

=== Associator model (full training set) ===

FPGrowth found 5 rules (displaying top 5)

1. [HotDogs=f]: 2 ==> [Buns=f]: 2    <conf:(1)> lift:(1.5) lev:(0.11) conv:(0.67)
2. [Chips=f]: 2 ==> [Coke=f]: 2    <conf:(1)> lift:(2) lev:(0.17) conv:(1)
3. [Ketchup=f]: 4 ==> [Buns=f]: 3    <conf:(0.75)> lift:(1.13) lev:(0.06) conv:(0.67)
4. [Buns=f]: 4 ==> [Ketchup=f]: 3    <conf:(0.75)> lift:(1.13) lev:(0.06) conv:(0.67)
5. [Coke=f]: 3 ==> [Chips=f]: 2    <conf:(0.67)> lift:(2) lev:(0.17) conv:(1)

|
```

Question – 8: Rule-Based Classification & Decision Tree Classification

Perform Prediction of Categorical Data using Rule-Based Classification and Decision Tree Classification through WEKA using any dataset. Compare the accuracy of both algorithms and plot the graph.

Aim

To classify categorical data using JRip (Rule-Based) and J48 (Decision Tree) algorithms in WEKA and compare their classification accuracy.

Steps

1. Open WEKA Explorer.
2. Load the weather.nominal.arff dataset in the Preprocess tab.
3. Verify that the class attribute is play.
4. Open the Classify tab.
5. Click Choose → rules → JRip.
6. Select Use training set and click Start.
7. Note the classification accuracy and generated rules.
8. Again click Choose → trees → J48.
9. Keep Use training set selected.
10. Click Start to generate the decision tree.
11. Compare the accuracy of JRip and J48.
12. Right-click the result → Visualize classifier errors to display the graph.

Output:

```
09:23:04 - rules.JRip

Time taken to build model: 0.02 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

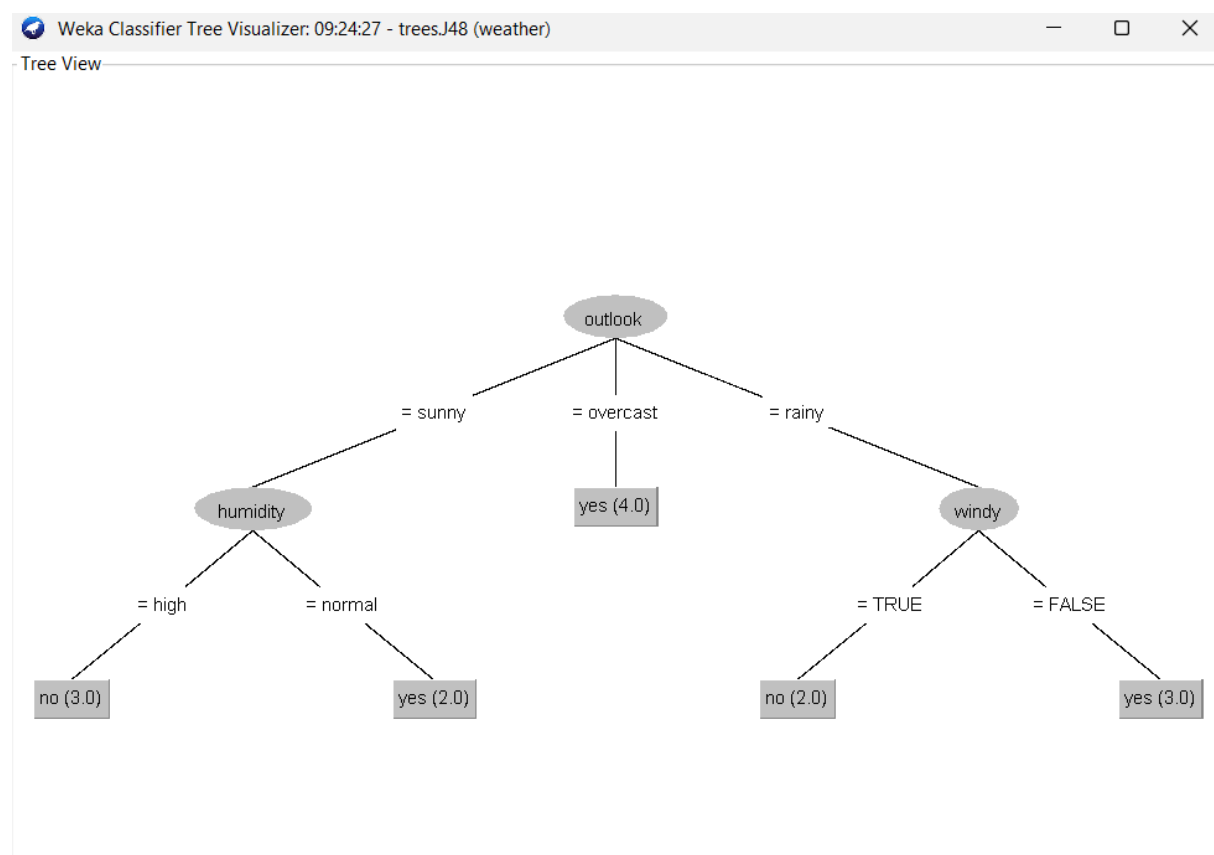
Correctly Classified Instances      14          100   %
Incorrectly Classified Instances    0           0   %
Kappa statistic                    1
Mean absolute error                 0
Root mean squared error            0
Relative absolute error             0   %
Root relative squared error        0   %
Total Number of Instances         14

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
      1.000    0.000    1.000     1.000    1.000     1.000    1.000    1.000    yes
      1.000    0.000    1.000     1.000    1.000     1.000    1.000    1.000    no
Weighted Avg.   1.000    0.000    1.000     1.000    1.000     1.000    1.000    1.000

=== Confusion Matrix ===

a b  <-- classified as
9 0 | a = yes
0 5 | b = no
```



Test options

☒ Use training set

☐ Supplied test set

☐ Cross-validation

☐ Percentage split

Set...

Folds10

%66

More options...

(Nom) play

StartStop

Result list (right-click for options)

09:23:04 - rulesJRip

09:24:27 - treesJ48

Classifier output

Decision tree

| windy = TRUE: no (2.0)

| windy = FALSE: yes (3.0)

Number of Leaves : 5

Size of the tree : 8

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	14	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0		
Root mean squared error	0		
Relative absolute error	0	%	
Root relative squared error	0	%	
Total Number of Instances	14		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	yes
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	no
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

a b <-- classified as

9 0 | a = yes

0 5 | b = no